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EXPT NO. - 3

TITLE - MERGE SORT

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In [2]: import random
import time
import matplotlib.pyplot as plt

def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr)//2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    return result + left[i:] + right[j:]

# Input from user
arr = list(map(int, input("Enter elements separated by space: ").split()))
print("Input array:", arr)

start_time = time.time()
sorted_arr = merge_sort(arr)
end_time = time.time()

print("Sorted array:", sorted_arr)
print("Time taken:", end_time - start_time, "seconds")
print("=====")

# For n>5000
n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)]
    start_time = time.time()
    merge_sort(arr)
    end_time = time.time()
    time_taken = end_time - start_time
    time_values.append(time_taken)
    print("Time for", n, "elements:", time_taken)

# Plot graph
plt.plot(n_values, time_values, color = 'black', marker='o')
plt.xlabel("Number of Elements (n)")
plt.ylabel("Time Taken (seconds)")
plt.title("Merge Sort Time Complexity")
plt.grid(True)
plt.show()
```

Input array: [40, 20, 50, 60, 90, 30, 70, 80, 10]
Sorted array: [10, 20, 30, 40, 50, 60, 70, 80, 90]
Time taken: 0.0006082057952880859 seconds

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Time for 5000 elements: 0.05364823341369629
Time for 6000 elements: 0.0434572696685791
Time for 7000 elements: 0.05595970153808594
Time for 8000 elements: 0.062273502349853516

